

Multimodal Large Language Models (MLLMs)

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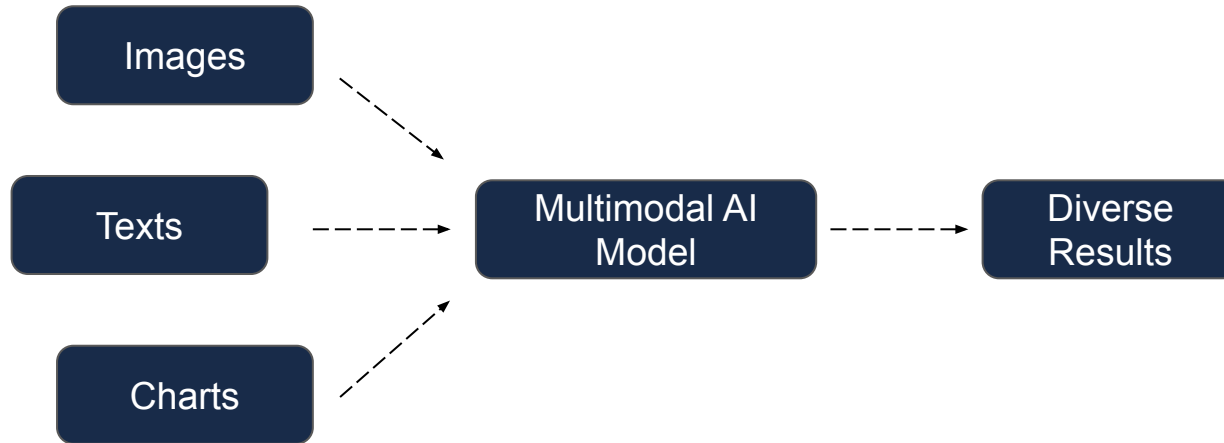
UC San Diego
JACOBS SCHOOL OF ENGINEERING

Single modality has limitations

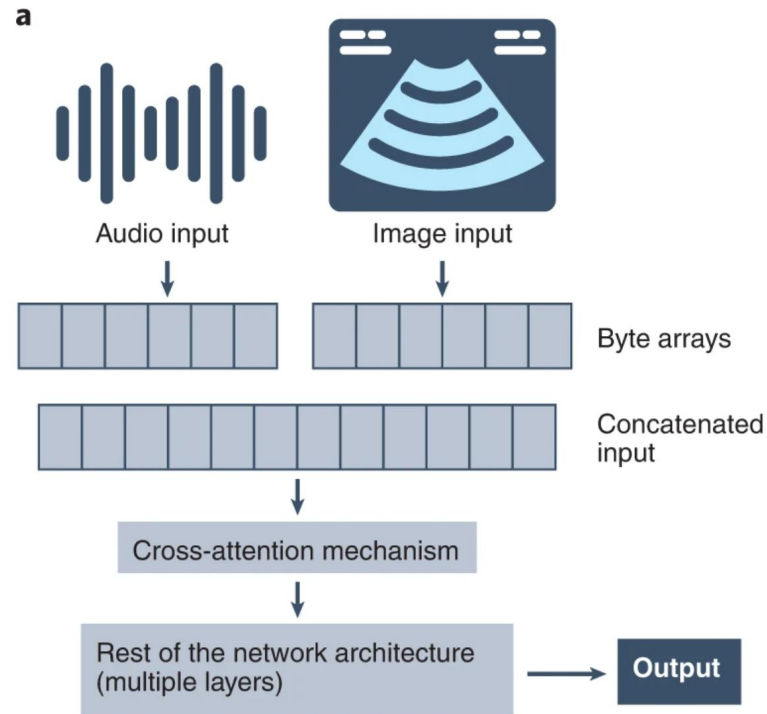




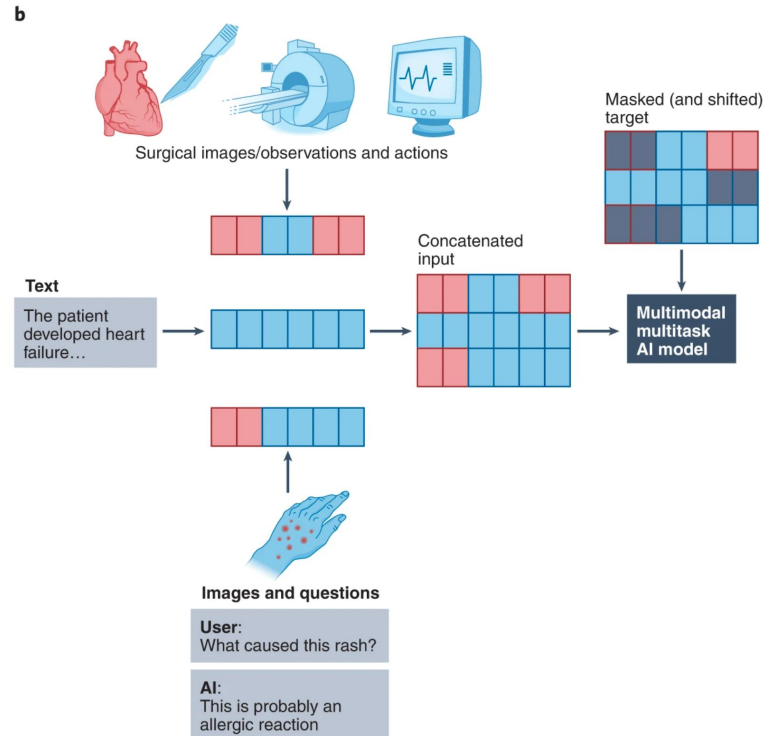
Multimodality Model



Example in Biomedical



Example in Biomedical



Examples from GPT4v

(2) Prompt:

Describe the pointed region in the image.

Method	Validation set										Test set									
	in.		near.		out.		overall		in.		near.		out.		overall		in.		near.	
	C	S	C	S	C	S	C	S	C	S	C	S	C	S	C	S	C	S	C	S
OSCAR	85.4	11.9	84.0	11.7	80.3	10.0	83.4	11.4	84.8	12.1	82.1	11.5	73.9	9.7	80.9	11.3	85.4	14.3	85.0	14.2
Human	84.4	14.3	85.0	14.2	95.7	14.0	87.1	14.2	80.6	15.0	84.6	14.7	91.6	14.2	85.3	14.6	92.2	12.9	97.8	12.6
VIVO	92.2	12.9	97.8	12.6	87.5	11.5	88.3	12.4	89.0	12.9	87.8	12.6	80.1	11.1	86.6	12.4	103.7	13.7	95.6	13.4
ViaVL	103.7	13.7	95.6	13.4	83.8	11.9	94.3	13.1	98.0	13.6	95.2	13.4	78.0	11.5	92.5	13.1	103.9	14.5	96.5	13.8
UPO	103.9	14.5	96.5	13.8	83.5	12.3	94.3	13.6	98.9	14.3	94.7	13.9	77.9	12.1	92.3	13.6	-	-	-	-
aiPLUG	-	-	-	-	-	-	114.8	14.8	-	-	-	-	-	-	-	-	-	-	-	-
SunVL	113.7	-	110.9	-	115.2	-	115.2	-	113.7	-	110.9	-	115.2	-	115.2	-	113.7	-	110.9	-
LEMON	118.0	15.4	116.3	15.1	120.2	14.5	117.3	15.0	112.8	15.2	115.5	15.1	110.1	13.7	114.3	14.9	123.2	15.0	121.5	15.3
UniversalCap	123.2	15.0	121.5	15.3	120.4	14.4	122.1	15.0	118.9	15.4	120.6	15.0	114.9	14.1	119.3	15.1	-	-	-	-
CoCa	-	-	-	-	-	-	122.4	15.5	-	-	-	-	-	-	-	-	-	-	-	-
GIT ₀	100.7	13.8	97.7	13.5	89.6	12.5	96.6	13.4	-	-	-	-	-	-	-	-	126.9	16.1	125.8	16.2
GIT ₁	107.7	14.9	107.8	14.5	102.5	13.7	106.9	14.4	-	-	-	-	-	-	-	-	124.2	16.4	125.5	16.1
GIT	129.8	16.3	124.1	16.0	127.1	15.7	125.5	16.0	122.4	16.2	123.9	16.0	122.0	15.7	123.4	15.9	122.3	15.6	124.8	16.1
GIT2	126.9	16.1	125.8	16.2	130.6	15.8	126.9	16.1	124.2	16.4	125.5	16.1	122.3	15.6	124.8	16.1	-	-	-	-

GPT-4V:

The highlighted red region in the image is a table cell that contains a numerical value of 122.3. It is located in the "Test set" section of the table, under the "out" column and the "GIT2" row. This value represents the overall performance metric for the GIT2 method on the test set, in the "out" category.

Prompt:

Localize each person in the image using bounding box. What is the image size of the input image?



GPT-4V:

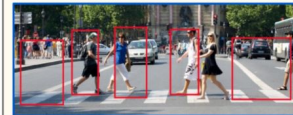
The image size of the input image is 1000 x 333 pixels.

Bounding boxes for each person:

- Person 1 (leftmost): (x1, y1) = (22, 108), (x2, y2) = (170, 300)
- Person 2: (x1, y1) = (200, 80), (x2, y2) = (290, 270)
- Person 3: (x1, y1) = (350, 70), (x2, y2) = (460, 280)
- Person 4: (x1, y1) = (540, 80), (x2, y2) = (640, 270)
- Person 5 (rightmost): (x1, y1) = (760, 100), (x2, y2) = (820, 290)

Note: These bounding box coordinates are approximate and may not be exact.

Visualization of Detected Bbox:



Prompt:

Which wisdom tooth is hardest to remove?



Note: Same query image as shown above, resized to save space

GPT-4V:

The lower wisdom teeth, especially those that are impacted, are typically considered the hardest to remove. This is because the lower jawbone is denser than the upper jawbone, and the impacted wisdom teeth may be more difficult to access. Additionally, the roots of the lower wisdom teeth are often longer and more curved than those of the upper wisdom teeth, making them more challenging to remove.

Prompt:

Count the number of apples in the image.



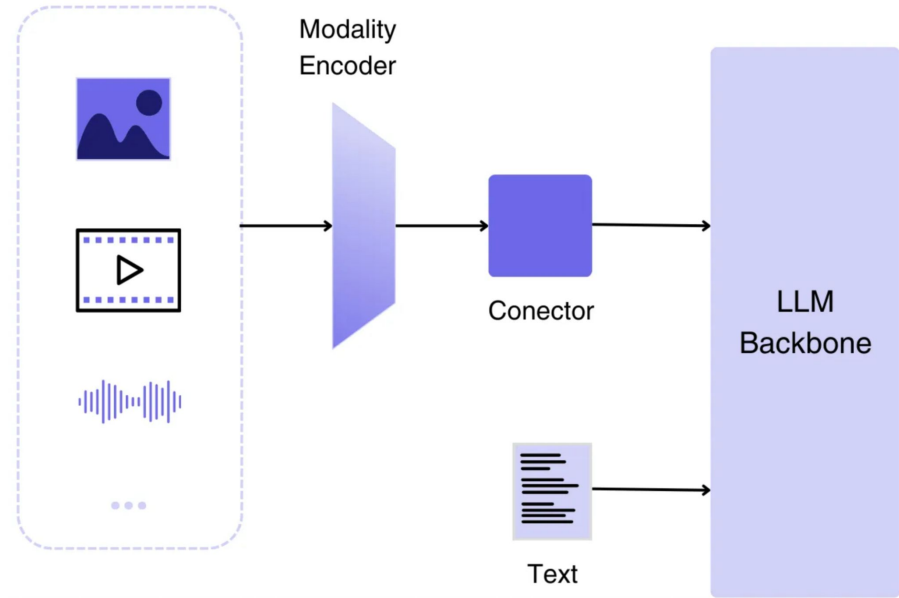
GPT-4V:

There are 7 apples in the image.



Fundamentals of Multimodal Training

- At high level, multimodal system consists
 - Encoder for each data modality to generate the embeddings
 - Way to align embeddings of different modalities into the same multimodal embedding space
 - A language model to generate text responses
 - Condition its responses on not just text, but also on other modalities





Early endeavors

- CLIP
- Flamingo
- Next-GPT



Early endeavors

- **CLIP**
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CLIP

- First model that could generalize to multiple image classification tasks with zero- and few-shot learning
- Map data of different modalities, text and images, into a shared embedding space
- Shared multimodal embedding space makes text-to-image and image-to-text tasks so much easier

Food101

guacamole (90.1%) Ranked 1 out of 101 labels



✓ a photo of **guacamole**, a type of food.

✗ a photo of **ceviche**, a type of food.

✗ a photo of **edamame**, a type of food.

✗ a photo of **tuna tartare**, a type of food.

✗ a photo of **hummus**, a type of food.

Youtube-BB

airplane, person (89.0%) Ranked 1 out of 23 labels



✓ a photo of a **airplane**.

✗ a photo of a **bird**.

✗ a photo of a **bear**.

✗ a photo of a **giraffe**.

✗ a photo of a **car**.

CLIP



Predetermined classes

Class 0	dog
Class 1	cat
...	
Class K	and

Classifier

Output probability

0.29	Class 0
0.52	Class 1
...	...
0.04	Class K

Final output

cat

Language model

Output probability

Token 1	Token 2	Token 3	
0.29	0.09	0.32	
0.52	0.11	0.21	...
...	...	...	
0.04	0.68	0.12	

Final output

cat **and** **dog** ...

CLIP



Several men in hard hats are operating a giant pulley system.

Workers look down from up above on a piece of equipment.

Two men working on a machine wearing hard hats.

Four men on top of a tall structure.

Three men on a large rig.



A black dog and a white dog with brown spots are staring at each other in the street.

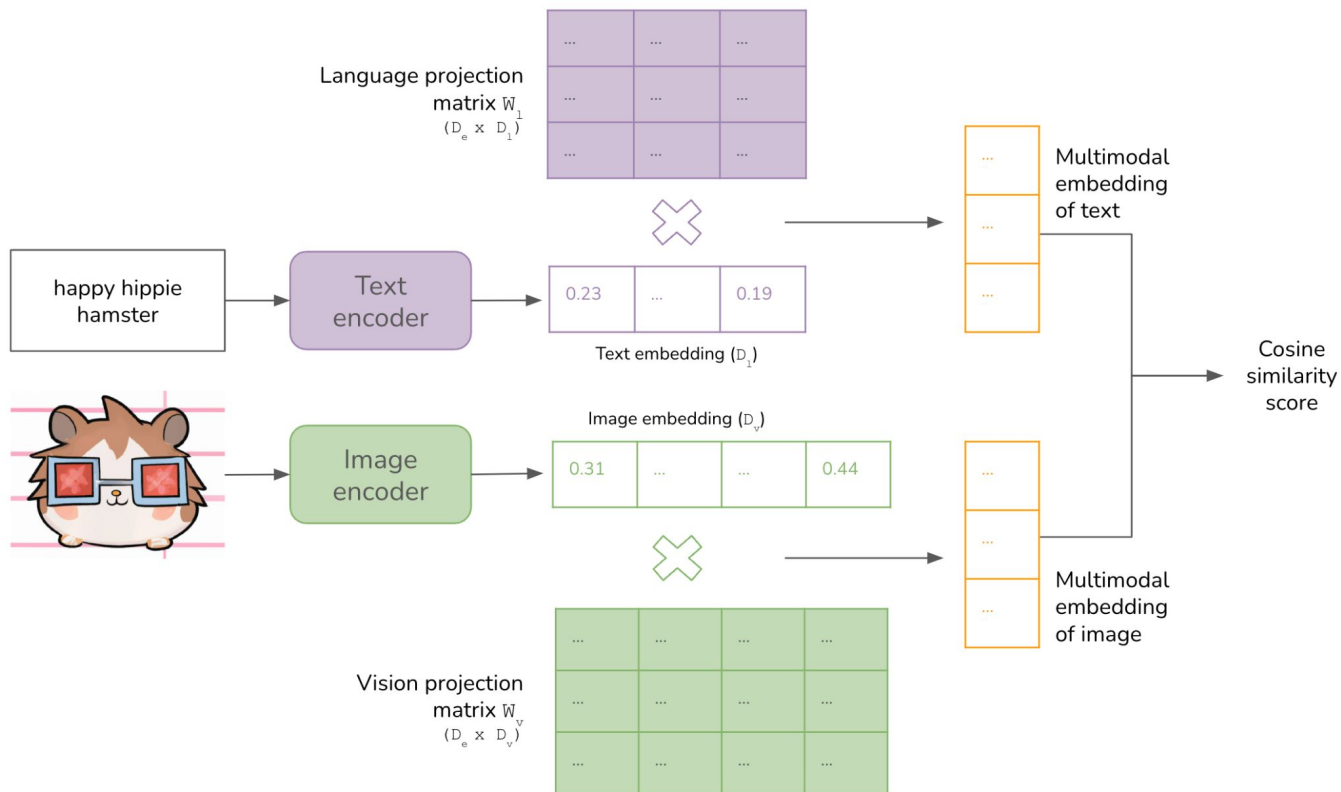
A black dog and a tri-colored dog playing with each other on the road.

Two dogs of different breeds looking at each other on the road.

Two dogs on pavement moving toward each other.

A black dog and a spotted dog are fighting.

CLIP



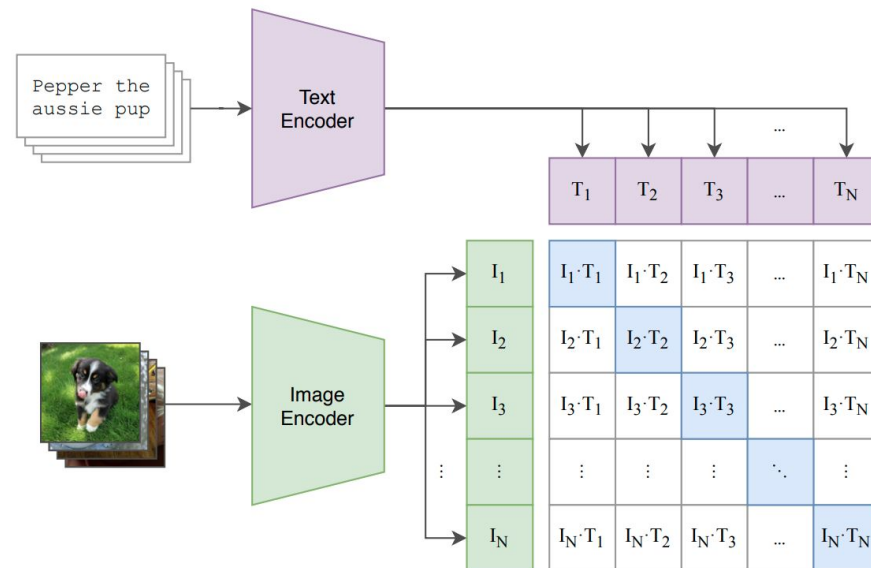
CLIP

- Instead of predicting the exact text of each image, CLIP predict whether a text is more likely to accompany an image than other texts
- For batch of $N(=32,768)$ pairs, minimize the two InfoNCE losses

$$\mathcal{L}_{\text{contrastive:txt2im}} = -\frac{1}{N} \sum_i \log\left(\frac{\exp(\mathbf{L}_i^T \mathbf{V}_i \beta)}{\sum_j \exp(\mathbf{L}_i^T \mathbf{V}_j \beta)}\right)$$

$$\mathcal{L}_{\text{contrastive:im2txt}} = -\frac{1}{N} \sum_i \log\left(\frac{\exp(\mathbf{V}_i^T \mathbf{L}_i \beta)}{\sum_j \exp(\mathbf{V}_i^T \mathbf{L}_j \beta)}\right)$$

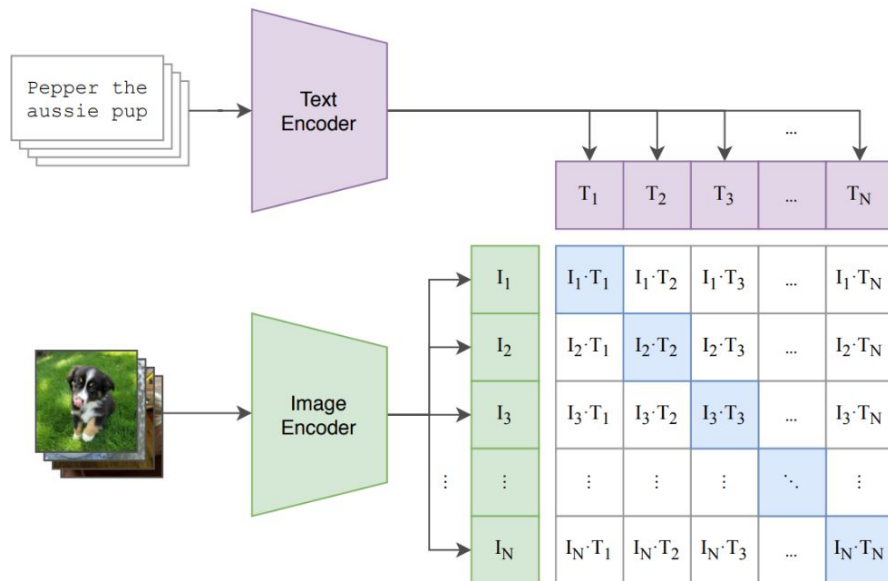
(1) Contrastive pre-training



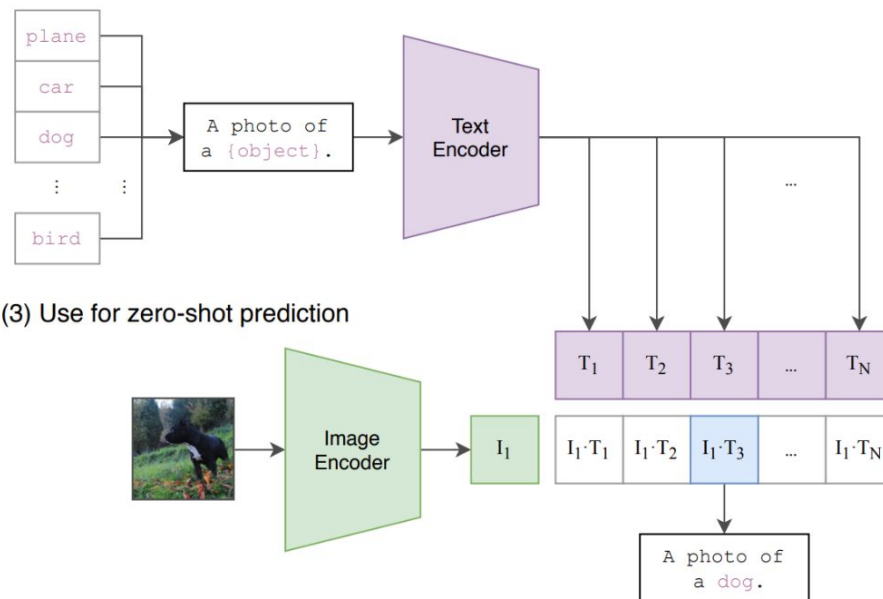
CLIP



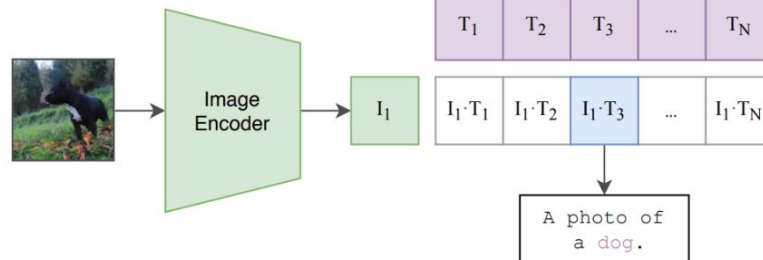
(1) Contrastive pre-training



(2) Create dataset classifier from label text



(3) Use for zero-shot prediction



CLIP



```
# image_encoder - ResNet or Vision Transformer
# text_encoder - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l] - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t - learned temperature parameter

# extract feature representations of each modality
I_f = image_encoder(I) #[n, d_i]
T_f = text_encoder(T) #[n, d_t]

# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)

# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)

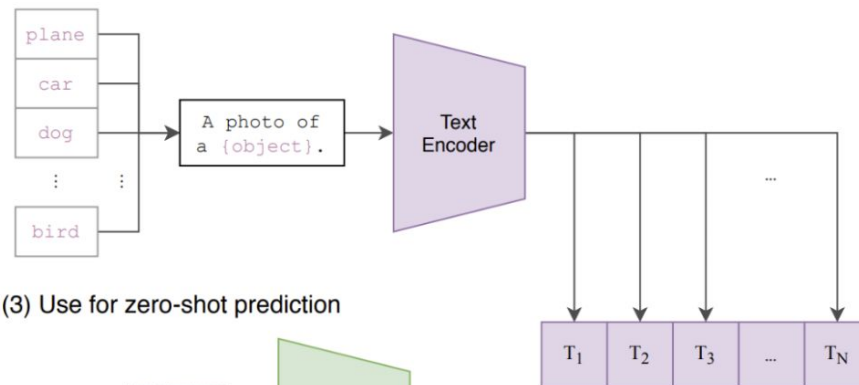
# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss = (loss_i + loss_t)/2
```

Figure 3. Numpy-like pseudocode for the core of an implementation of CLIP.

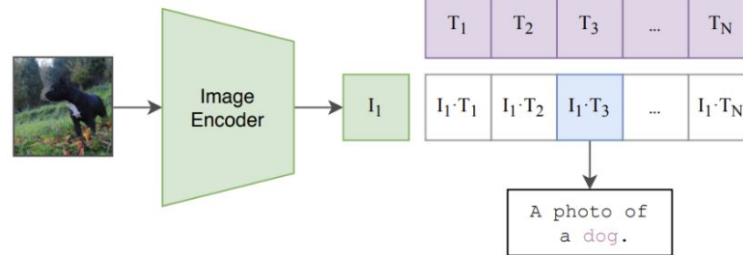
Problems with CLIP

- CLIP is not for generation
- Few-shot requires linear probing
- Essentially not a LLM
 - Cannot do many tasks, e.g. VQA
- Solutions?
 - Attach a LLM to it

(2) Create dataset classifier from label text



(3) Use for zero-shot prediction



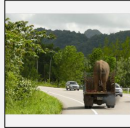
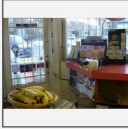
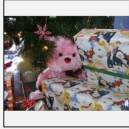


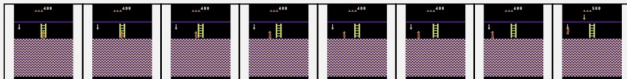
Early endeavors

- CLIP
- **Flamingo**
- Next-GPT

Flamingo



Input Prompt			
	Question: What do you think the capacities of these are? Answer:	Question: What is odd about this image? Explain why it is unusual. Answer:	Question: What country is this? Why do you think so? Answer:
Completion	The floppy disk is 1.44MB and the CD is 700MB.	The image is odd because the elephant is in the back of the truck. It is unusual because elephants are not usually transported in the back of a truck.	It is Canada. I think so because the flag is the Canadian flag.
Input Prompt			
	Question: What nutrients is the fruit in the image rich in? Answer:	Question: What is the purpose of this? Answer:	Question: What season is this? Explain why. Answer:
Completion	Vitamin C, potassium, and fiber.	To keep the mosquitoes out of the bed.	This is winter season. The Christmas tree and the presents are the main reasons why it is winter.

Input Prompt								Completion
								The dachshund puppy is being weighed on a scale.
Question: What is happening here? Answer:								
								A sword.
Question: What object is the avatar picking up? Answer:								
								The object is a small plastic bowl.
Question: What is the object being lifted? Answer:								
								A scissors, a pen, and a rubber band.
Question: What are the three objects in this video? Answer:								
								Flamingo.
Question: What is written here? Answer:								
								he falls down.
What happens to the man after hitting the ball? Answer:								

Flamingo



P This is a dog. It's a white fluffy dog.

What season is this?

P It looks like a summer day.

Why do you think so?

P The grass is green and the dog is not wearing a sweater.

What breed is the dog?

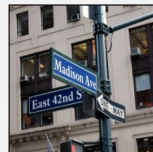
P It's a Samoyed.

Is the dog running?

P No, it's sitting.

Can you describe the pose of its back legs?

P The back legs are bent and the dog is sitting on its haunches.



Which city is this?

P This is a street sign in New York City.

Which street corner is it?

P It is the intersection of Madison Avenue and East 42nd Street.

Which subway station is nearby?

P The closest subway station is Grand Central Station.

How about the nearest park?

P The nearest park is Bryant Park.

There is another sign. What does it say?

P It says "One way".

I would like you to read the color and then say the color it is written in. For example:

PURPLE

Color is "Purple" and it is written in red.

GREEN

Color is "Green" and it is written in blue.

YELLOW

Color is "Yellow" and it is written in green.

BLACK

Color is "Black" and it is written in yellow.

Well done! Do you know the name of the test these images come from?

P I think it is called the Stroop test.

Can you explain how humans perform in this test?

P Humans are slower when the color of the word and the color of the word are different.

How about you?

P I am not affected by this difference.



What is the difference between these two images?

P The first one is a triangle and the second one is a hexagon.

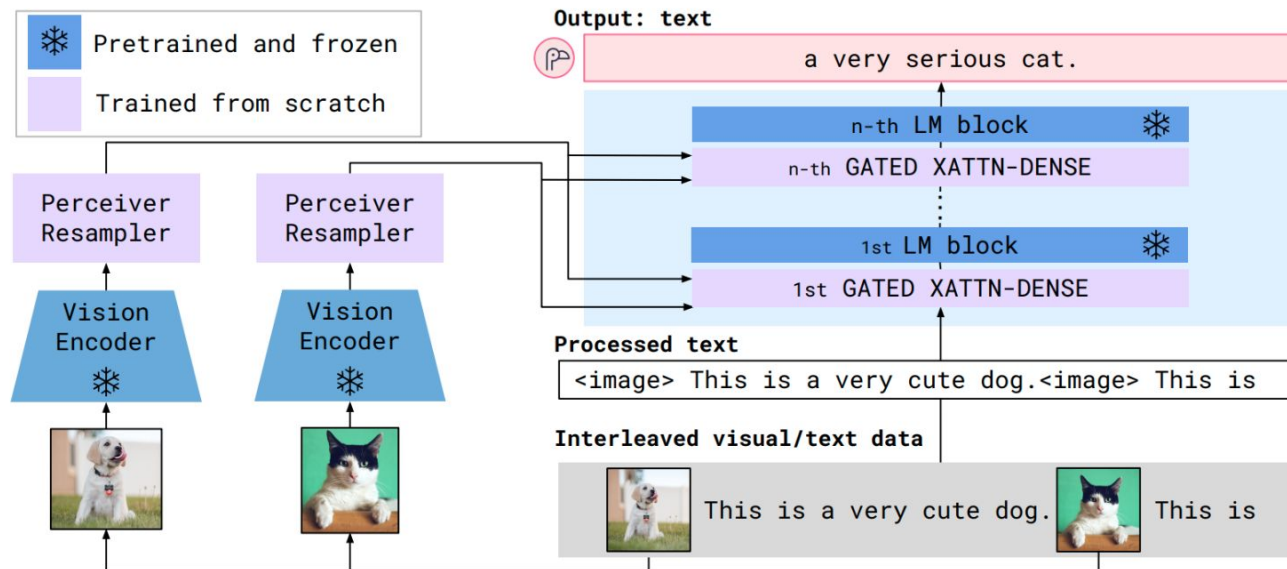
Which one has larger internal angles?

P The hexagon has larger internal angles.

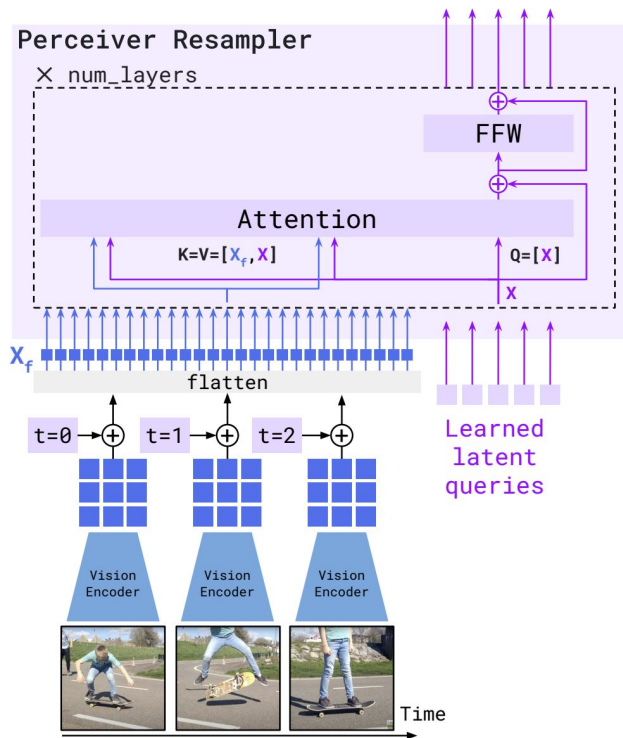
What color are they?

P They are both red.

Flamingo



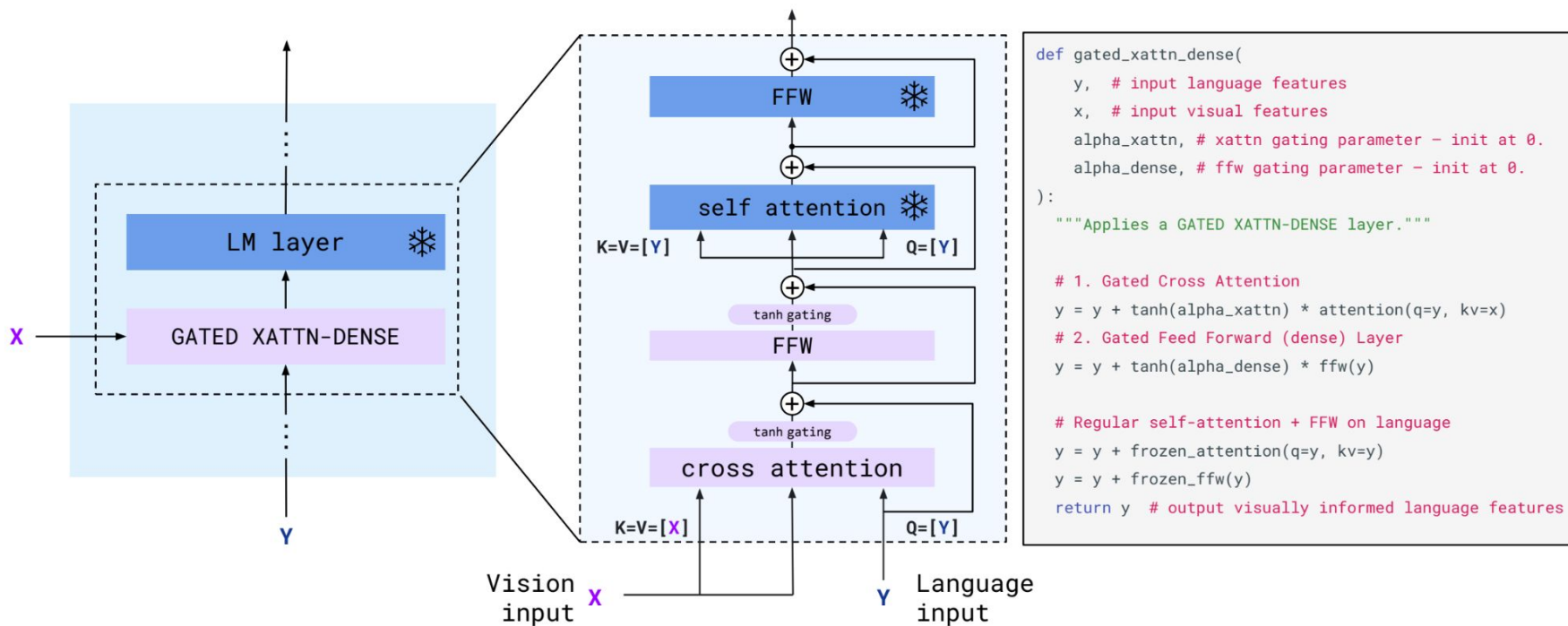
Flamingo



```
def perceiver_resampler(
    x_f, # The [T, S, d] visual features (T=time, S=space)
    time_embeddings, # The [T, 1, d] time pos embeddings.
    x, # R learned latents of shape [R, d]
    num_layers, # Number of layers
):
    """The Perceiver Resampler model."""

    # Add the time position embeddings and flatten.
    x_f = x_f + time_embeddings
    x_f = flatten(x_f) # [T, S, d] -> [T * S, d]
    # Apply the Perceiver Resampler layers.
    for i in range(num_layers):
        # Attention.
        x = x + attention_i(q=x, kv=concat([x_f, x]))
        # Feed forward.
        x = x + ffw_i(x)
    return x
```

Flamingo



Flamingo

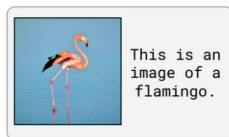
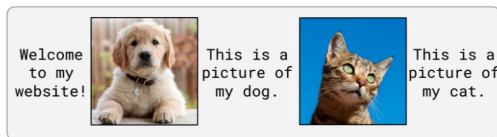


Image-Text Pairs dataset



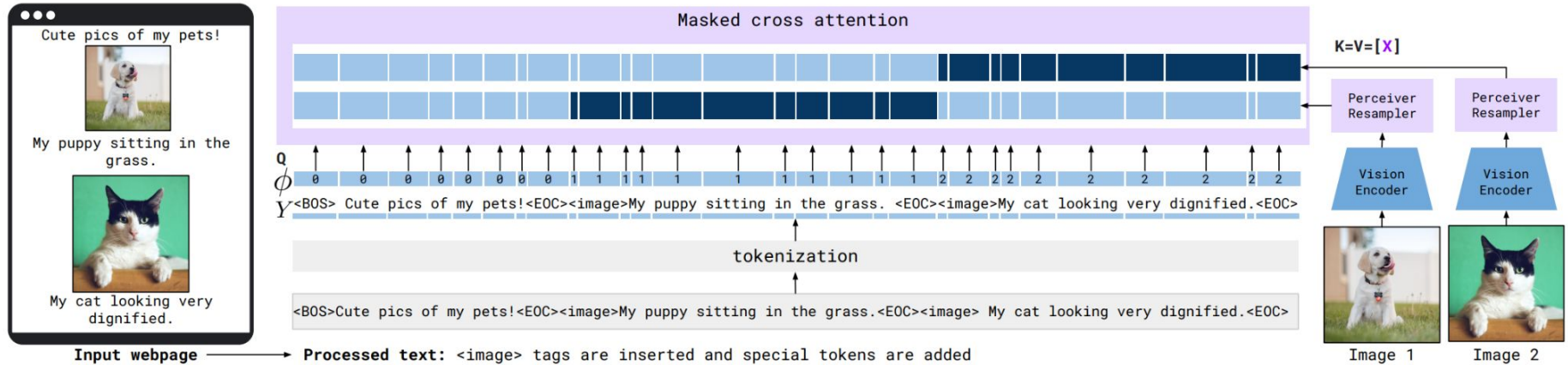
Video-Text Pairs dataset



Multi-Modal Massive Web (M3W) dataset

Dataset	Type	Size	How	Training weight
M3W	Interleaved image and text dataset	43M webpages	For each webpage, they sample a random subsequence of 256 tokens and take up to the first 5 images included in the sampled sequence.	1.0
ALIGN	(Image, text) pairs	1.8B pairs	Texts are alt-texts, averaging 12 tokens/text.	0.2
LTIP	(Image, text) pairs	312M pairs	Texts are long descriptions, averaging 20.5 tokens/text.	0.2
VTP	(Video, text) pairs	27M short videos	~22 seconds/video on average	0.03

Flamingo



Flamingo



Vision to Text tasks (input=vision, output=text)

Support examples

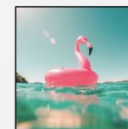
Query



A cat wearing sunglasses.



Elephants walking in the savanna.



<BOS><image>Output: A cat wearing sunglasses.<EOC><image>Output: Elephants walking in the savanna.<EOC><image>Output:

Processed prompt

Visual Question Answering Task (input=vision+text, output=text)

Support examples

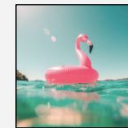
Query



What's the cat wearing? sunglasses



How many animals? 3



What is on the water?

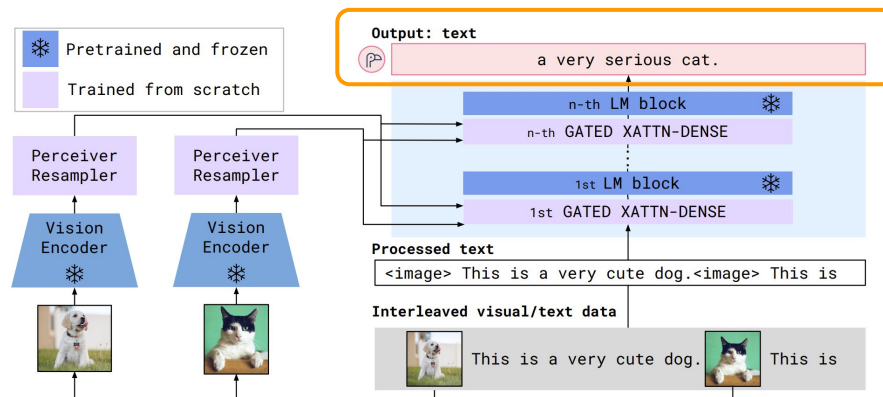
<BOS><image>Question: What's the cat wearing? Answer: sunglasses<EOC><image>Question: How many animals? Answer: 3<EOC><image>Question: What is on the water? Answer:

Processed prompt



Flamingo Only Outputs Text

- Encoder + LLM $\rightarrow$ text output
- Only conditioned on visual input and text
- Can we extend to other modality output?

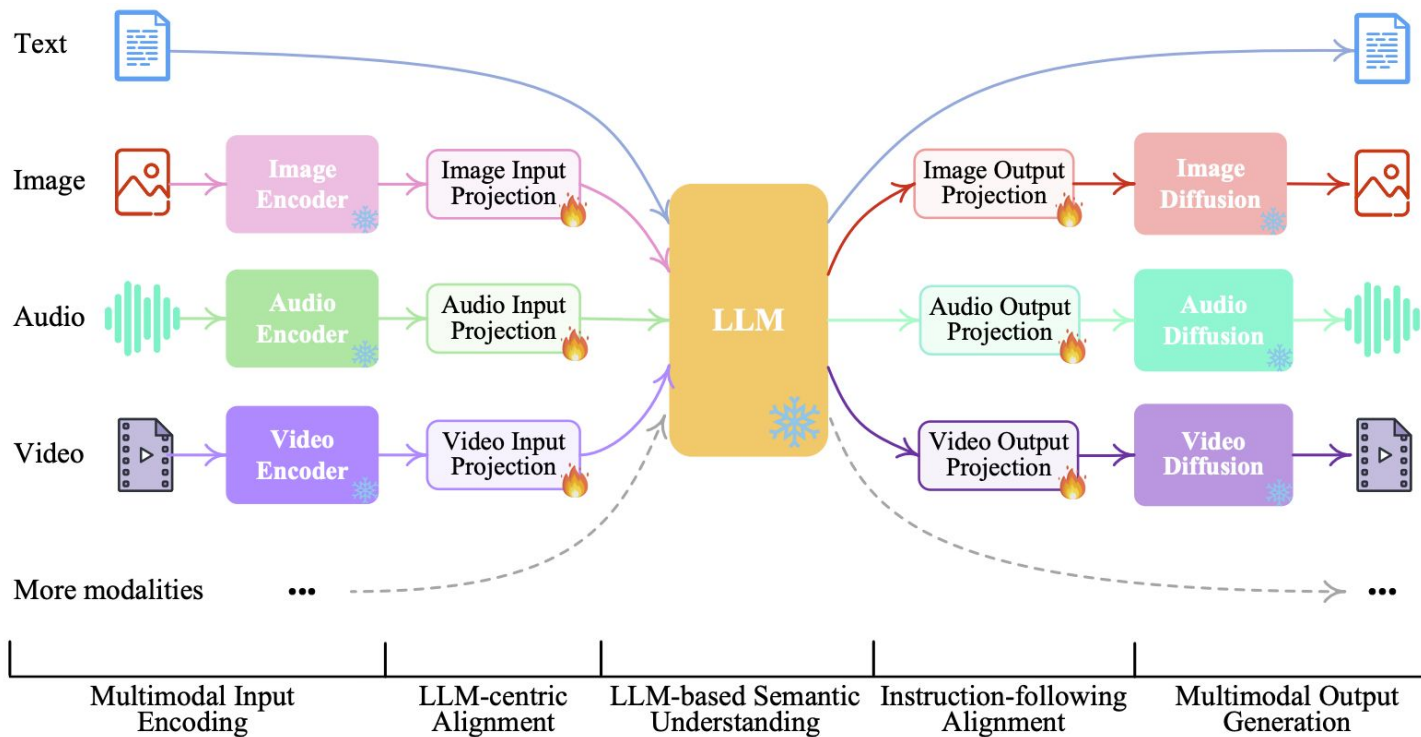




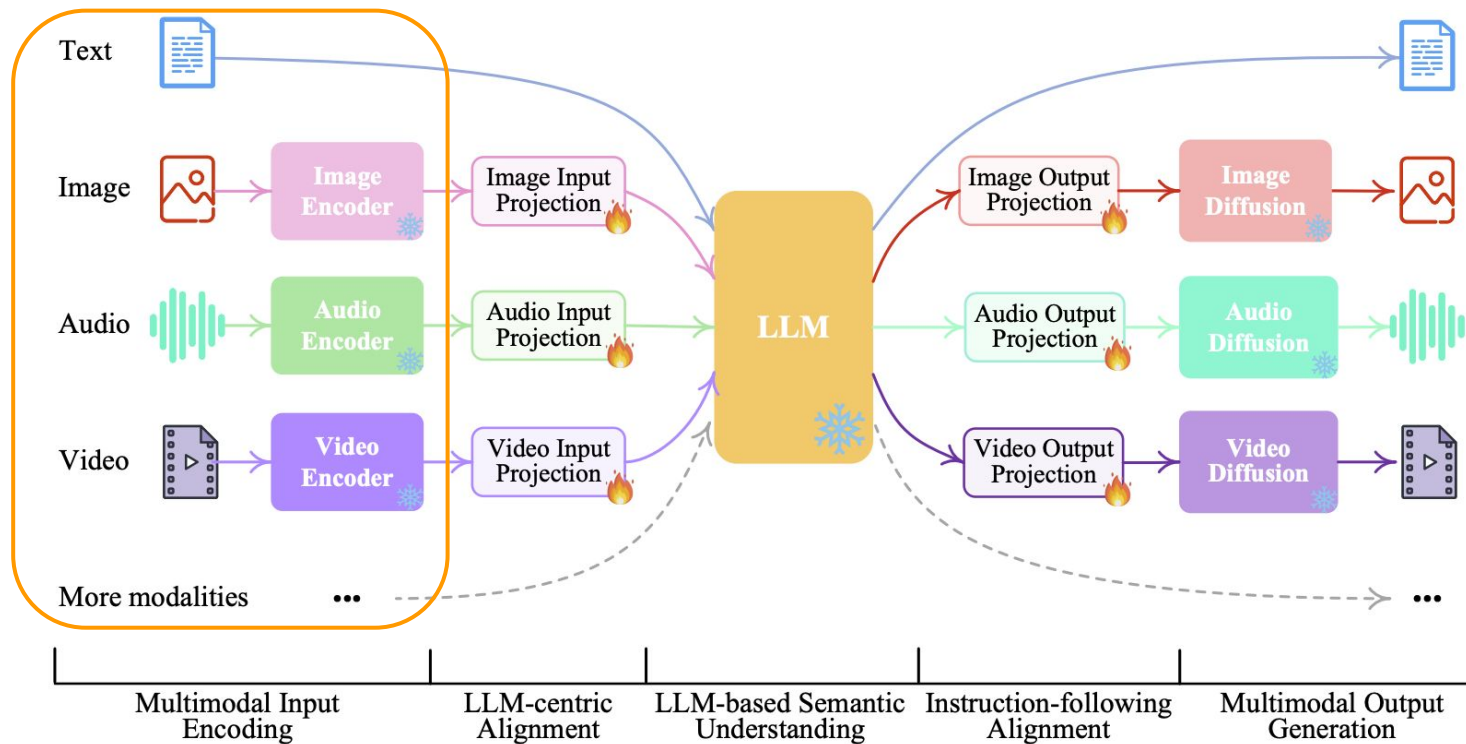
Early endeavors

- CLIP
- Flamingo
- **Next-GPT**

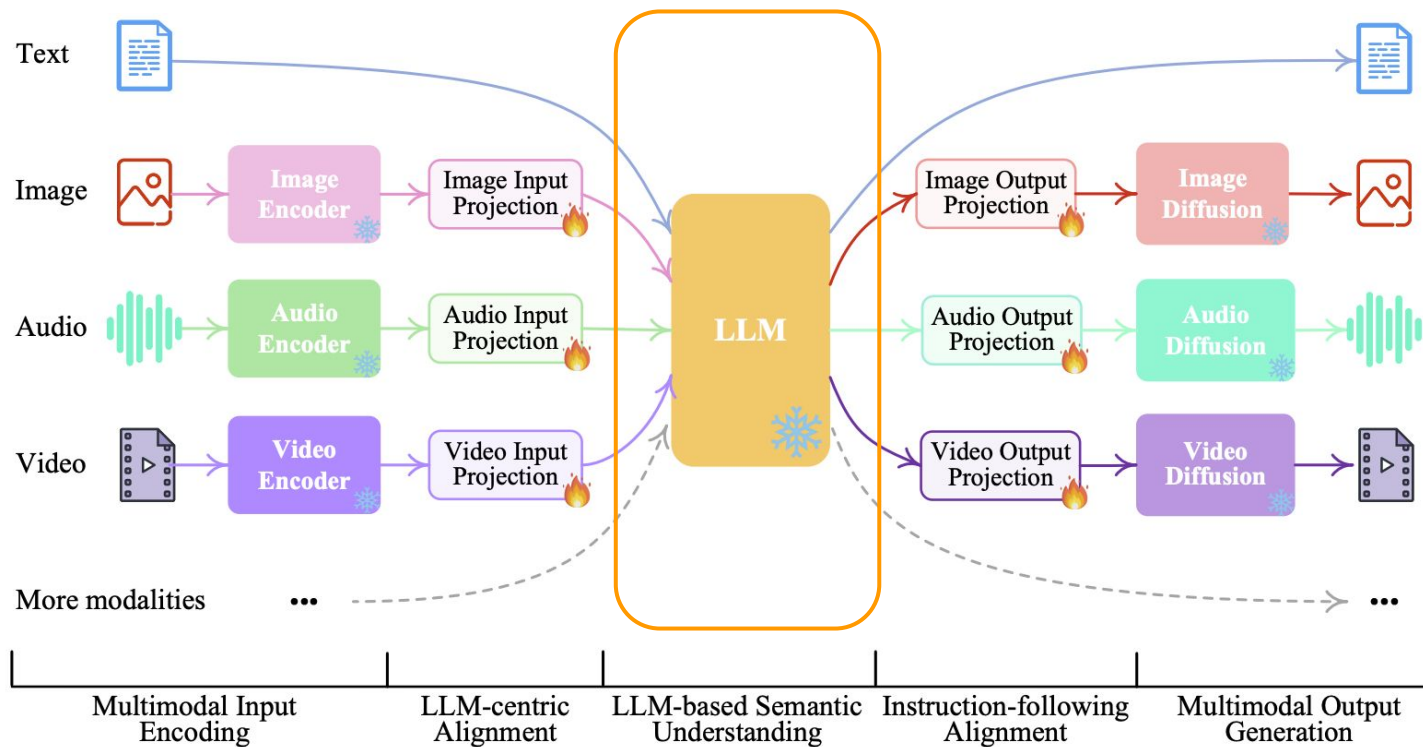
Next-GPT



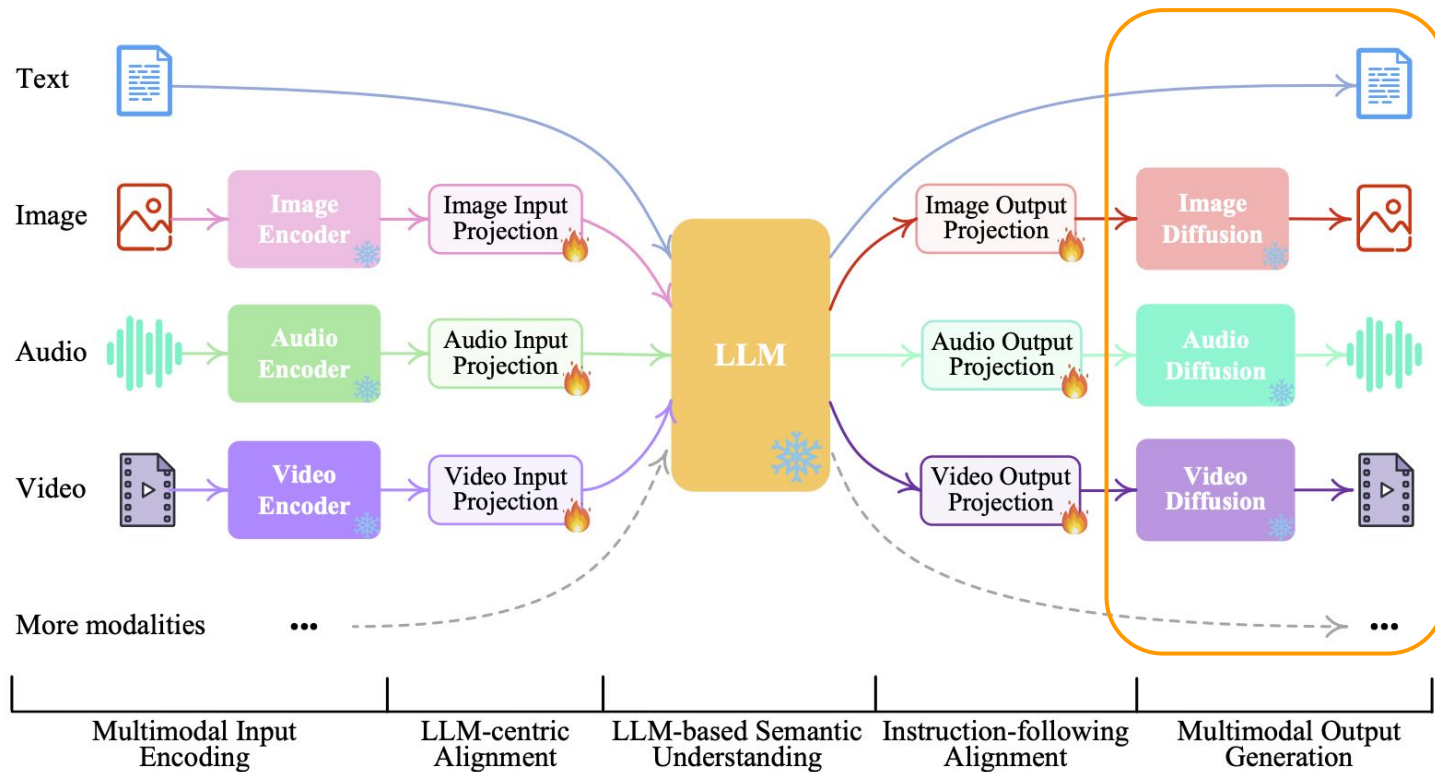
Next-GPT



Next-GPT



Next-GPT





Next-GPT

Table 1. Summary of NExT-GPT system configuration. Only 1% of parameters need updating during fine-tuning.

	Encoder		Input Projection		LLM		Output Projection		Diffusion	
	Name	Param	Name	Param	Name	Param	Name	Param	Name	Param
Text	—	—	—	—	Vicuna (LoRA	7B❄️ 33M🔥)	—	—	—	—
Image	ImageBind	1.2B❄️	Grouping	28M🔥			Transformer	31M🔥	SD	1.3B❄️
Audio							Transformer	31M🔥	AudioLDM	975M❄️
Video							Transformer	32M🔥	Zeroscope	1.8B❄️



Web Image-Text



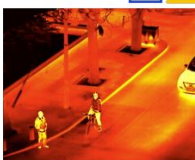
Depth Sensor Data



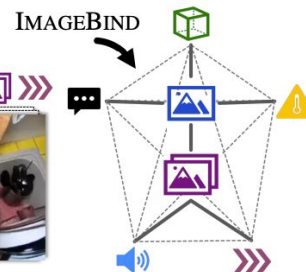
Web Videos



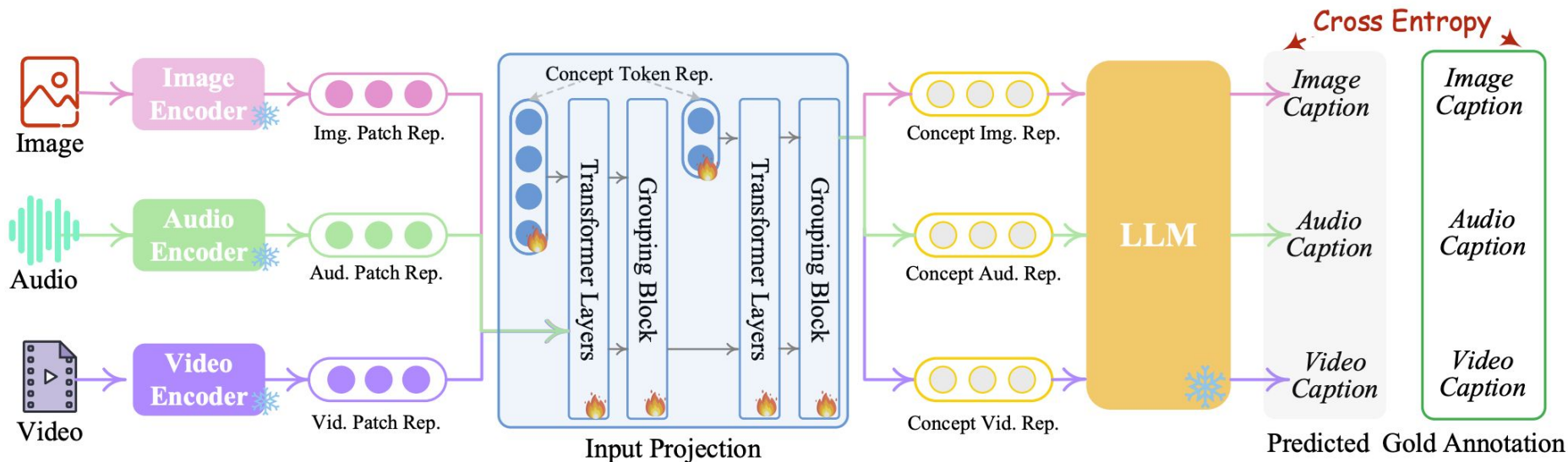
Thermal Data



Egocentric Videos

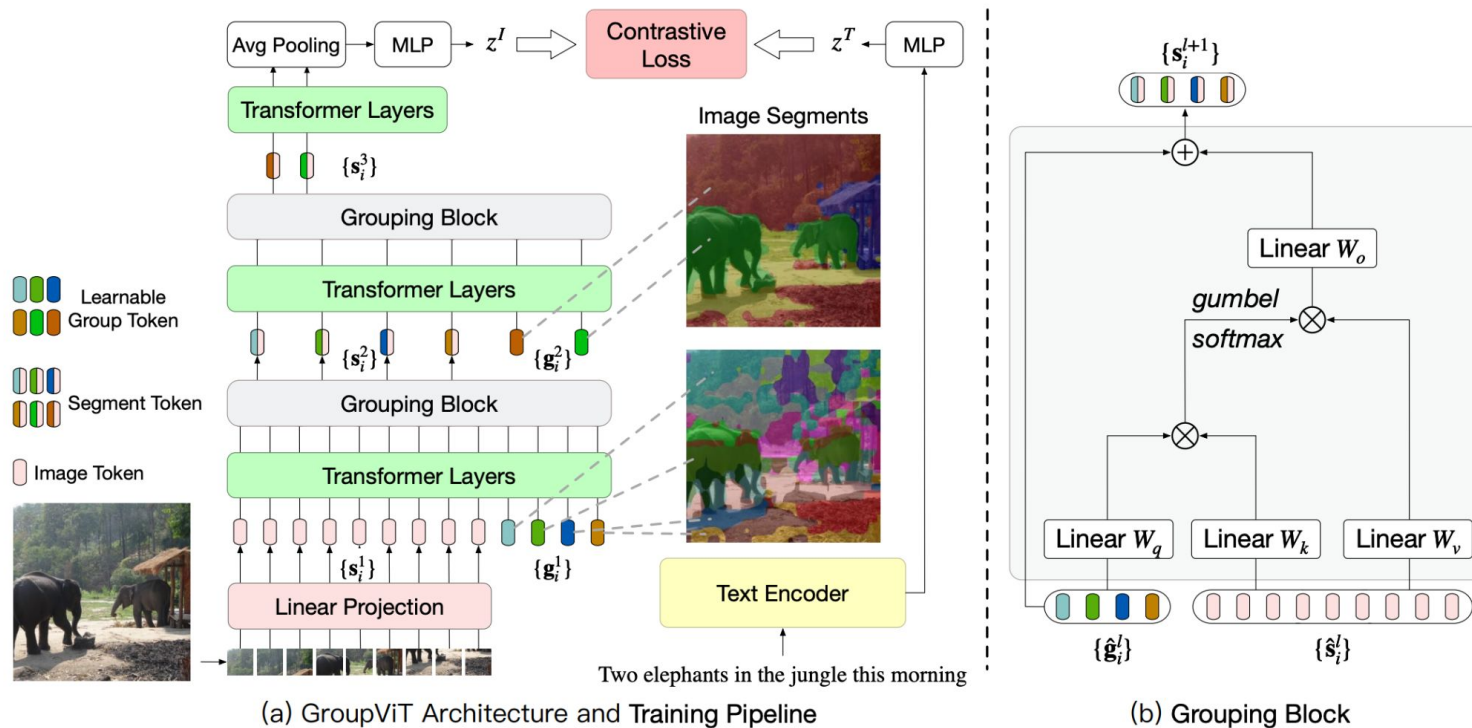


Next-GPT

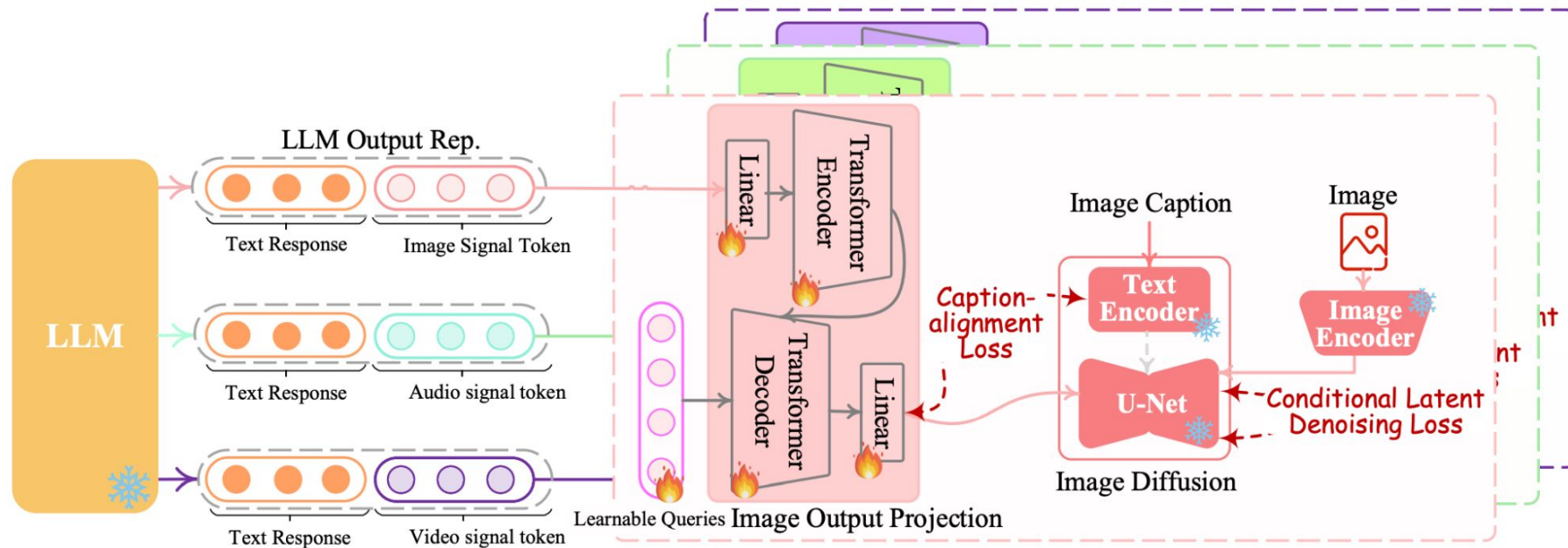


(a) Encoding-side LLM-centric Alignment

Next-GPT

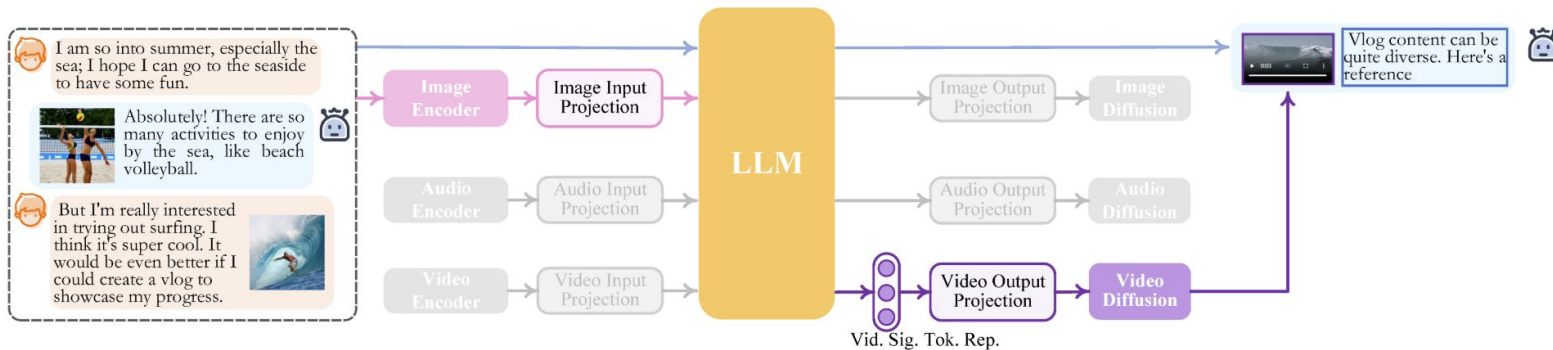
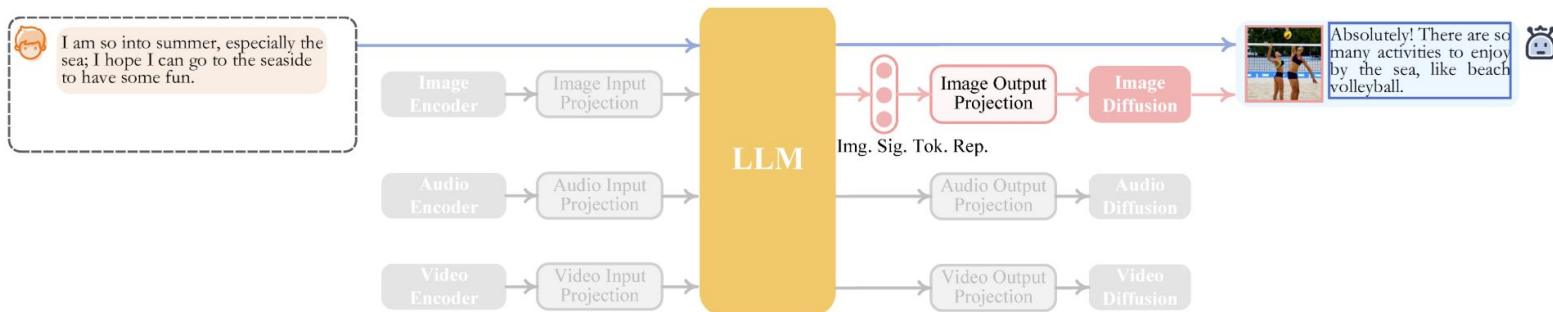


Next-GPT

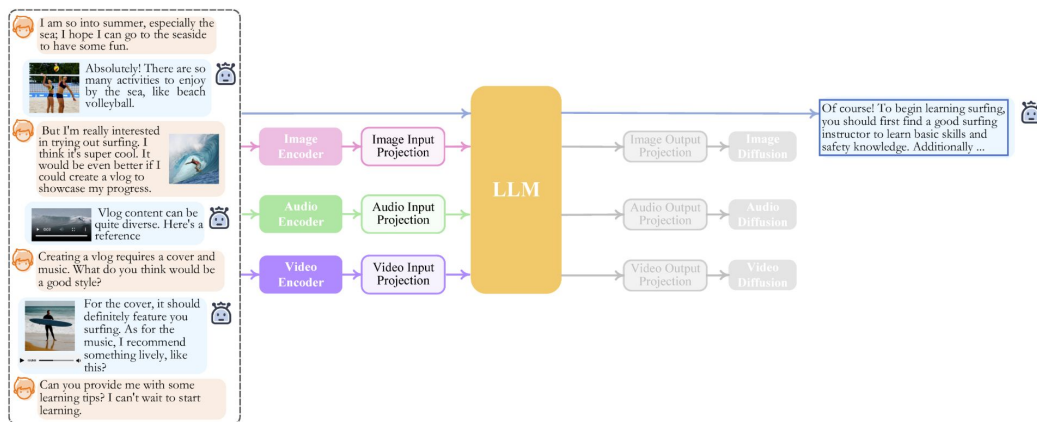
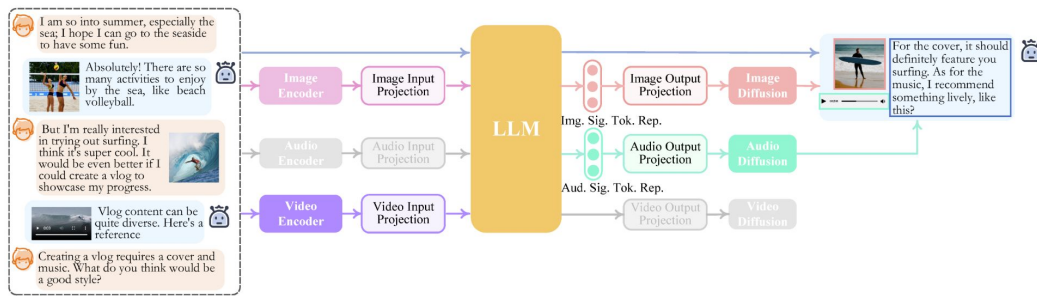


(b) Decoding-side Instruction-following Alignment

Next-GPT

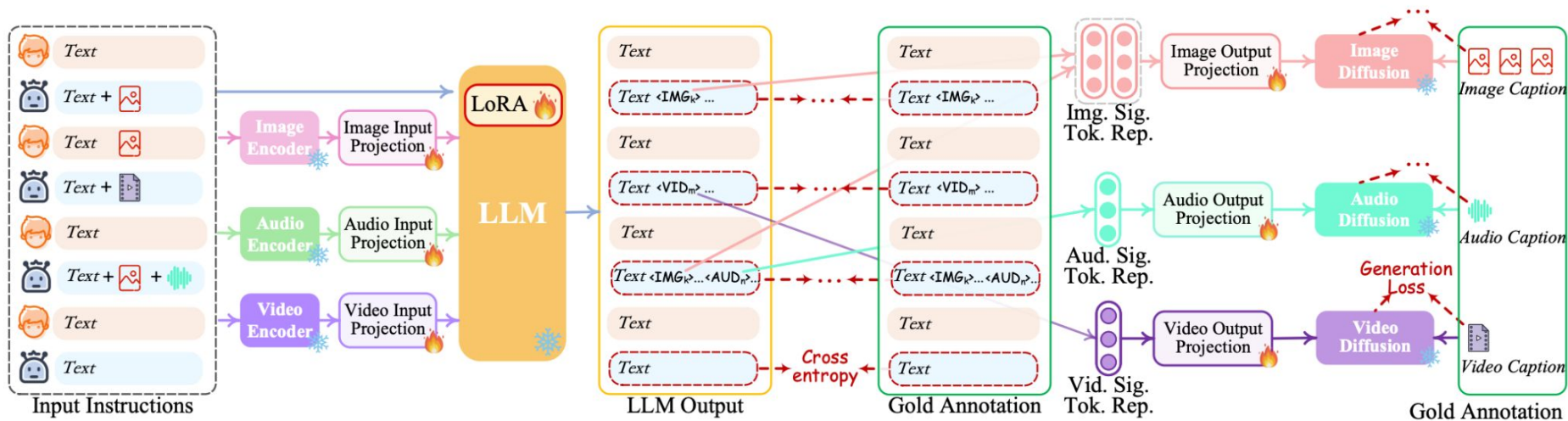


Next-GPT





Next-GPT – MosIT (Modality Switching)



More to come

- Chameleon
- Transfusion



More to come

- **Chameleon**
- Transfusion

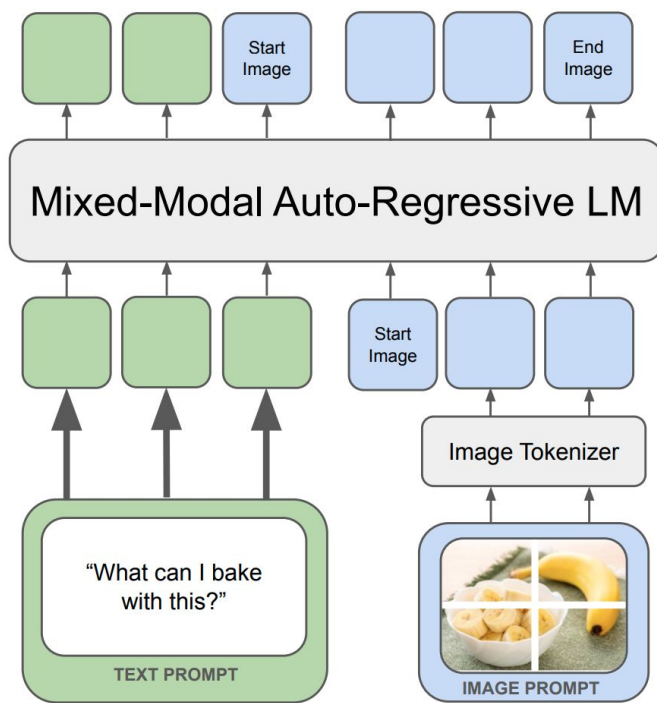




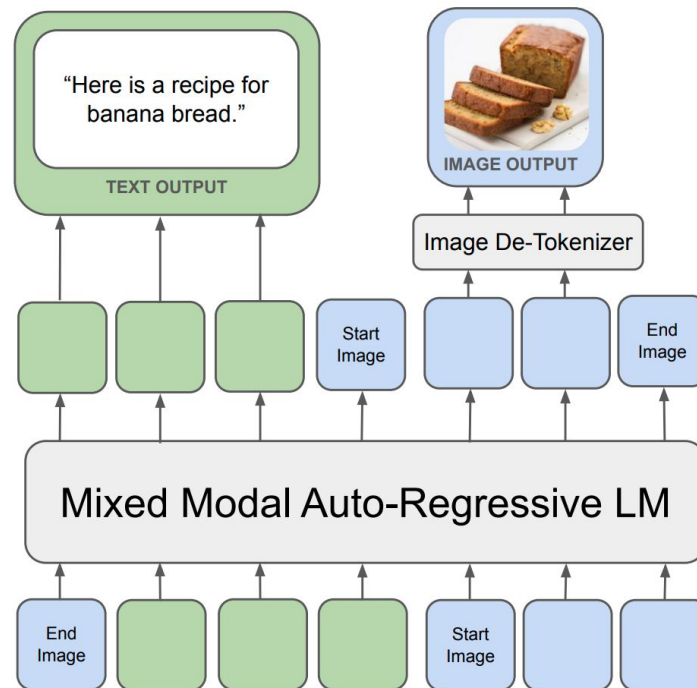
Chameleon

- Overcoming Modality Separation
 - all modalities are projected into a shared representational space from the start
- Unified Reasoning and Generation
 - capable of reasoning over and generating interleaved image-text documents
- Addressing Optimization Challenges

Chameleon – Method



(a) Mixed-Modal Pre-Training



(b) Mixed-Modal Generation



Chameleon – PT

- Pretraining
 - Image tokenization: encodes 512x512 image to 1024 discrete tokens from a codebook of size 8192
 - Tokenizer: Trained a BPE tokenizer, with a vocabulary size of 65,536 including image codebook tokens
 - Two stage pretraining data split
 - First stage: 80% of data
 - Second stage: last 20%



Chameleon – PT

- Pretraining
 - First stage
 - Text-only: pre-training data used to train LLaMa-2 and CodeLLaMa, total 2.9 trillion tokens
 - Text/Image Interleaved: 400 billion tokens of interleaved text and image data from public source (not directly mentioned)
 - Second stage
 - Lower the weight of the first stage data by 50% and mix in higher quality datasets
 - Include a filtered subset of the train sets from a large collection of instruction tuning sets



Chameleon – PT

- Optimization Stability

- Architecture largely follows Llama-2

- divergence to the softmax operation
- each modality will try to “compete” with the other by increasing its norms slightly
 - logit drift problem in unimodal setting

- **Chameleon-34B:** $h = x + \text{attention_norm}(\text{attention}(x))$
output = $h + \text{ffn_norm}(\text{feed_forward}(h))$

- Llama2:** $h = x + \text{attention}(\text{attention_norm}(x))$
output = $h + \text{feed_forward}(\text{ffn_norm}(h))$



Chameleon – PT

Model	Params	Context Length	GQA	Tokens	LR	Epochs	Dropout	Zloss	Qknorm
LLaMa-1	7B	2k	×	1.0T	3.0×10^{-4}	1.0	0.0	0.0	×
	33B	2k	×	1.4T	1.5×10^{-4}	1.0	0.0	0.0	×
LLaMa-2	7B	4k	×	2.0T	3.0×10^{-4}	1.0	0.0	0.0	×
	34B	4k	✓	2.0T	1.5×10^{-4}	1.0	0.0	0.0	×
Chameleon	7B	4k	×	4.4T	1.0×10^{-4}	2.1	0.1	10^{-5}	✓
	34B	4k	✓	4.4T	1.0×10^{-4}	2.1	0.0	10^{-5}	✓

Chameleon	Concurrent GPUs	GPU Hours
7B	1024	856481
34B	3072	4282407



Chameleon – SFT

- SFT
 - Text
 - Code
 - Visual Chat
 - Image Generation
 - Interleaved Text/Image Generation
 - Safety

More to come

- Chameleon
- **Transfusion**



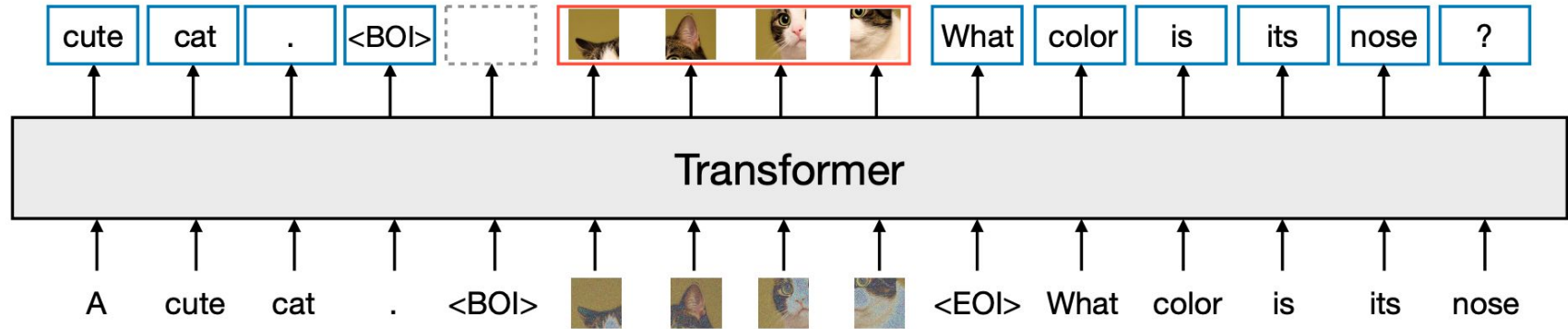


Transfusion (ICLR 2025 Oral)

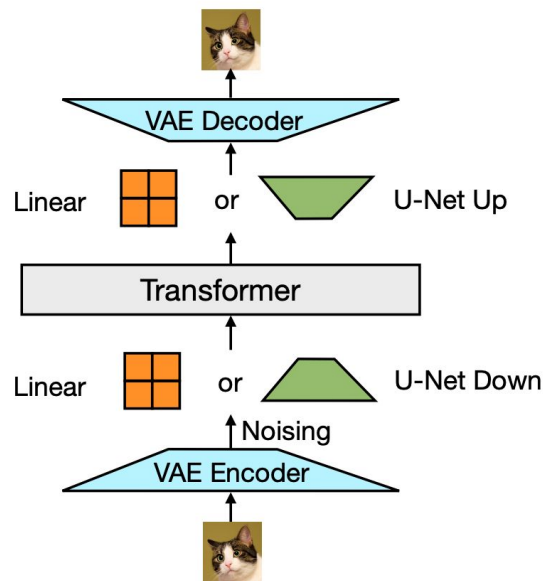
- Unified Multi-Modal Learning
 - Single model without discretizing the image
- Combining Strengths of Different Losses
 - LM loss for text gen and DDPM loss for image
- Scalability
 - achieves approximately 2× lower FID scores than Chameleon models
 - more efficient at learning text-to-text prediction



Transfusion – Method

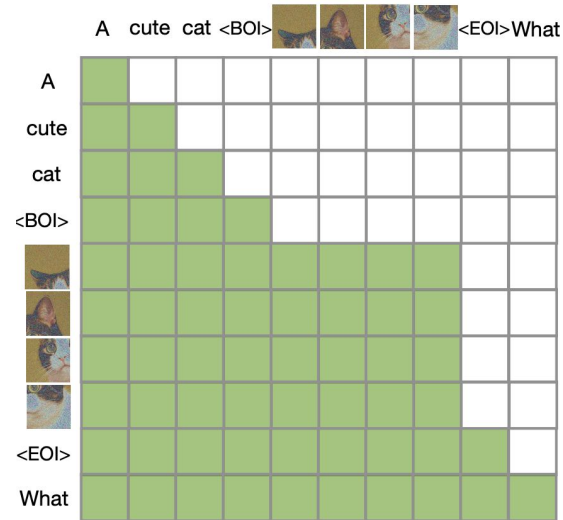


Transfusion – architecture



- Majority belongs to single Transformer
- lightweight modality-specific components with unshared parameters for text and image
 - Text: embedding matrices
 - Image: 1) linear layer, 2) U-Net

Transfusion – Method



- Text causal attention
- Image bidirectional attention



Transfusion – Method

- Training objective:
 - $\mathcal{L}_{\text{Transfusion}} = \mathcal{L}_{\text{LM}} + \lambda \cdot \mathcal{L}_{\text{DDPM}}$
 - language modeling objective to predictions of text tokens
 - diffusion objective to predictions of image patches
 - add noise to each input latent image according to the diffusion process before patchification



Transfusion – Method

- Inference:
 - Switches between LM and diffusion (On BOI and EOI)
 - For sampling images:
 - Append pure noise in the form of n image patches
 - Denoise over step T



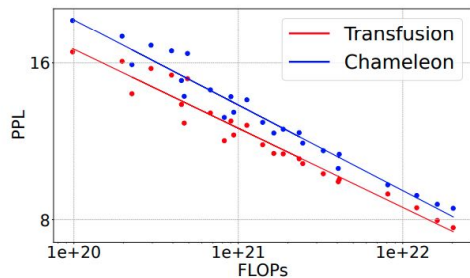
Transfusion – Evaluations

- Training data:
 - Similar to Chameleon on text (2T text tokens)
 - For image, 380M licensed Shutterstock images and captions
 - Each center-cropped and resized to 256×256
 - sample 0.5T tokens (patches) from two datasets at a 1:1
- Model sizes:
 - 0.16B, 0.37B, 0.76B, 1.4B, and 7B

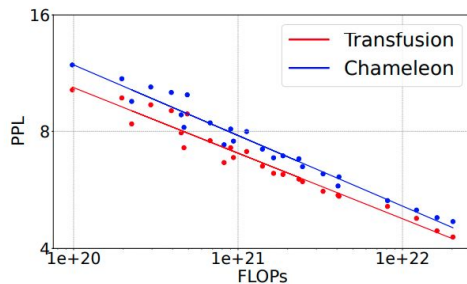
Size	Layers	Emb Dim	Att Heads
0.16B	16	768	12
0.37B	24	1024	16
0.76B	24	1536	24
1.4B	24	2048	16
7B	32	4096	32



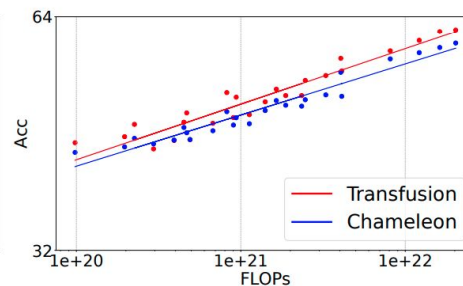
Transfusion – Evaluations



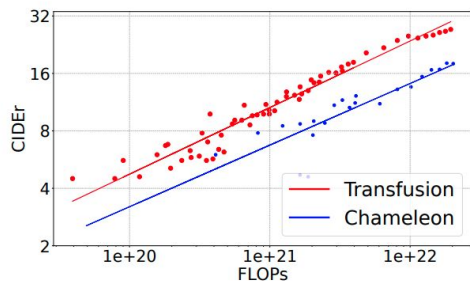
C4 Perplexity



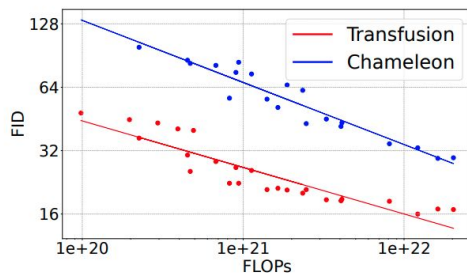
Wikipedia Perplexity



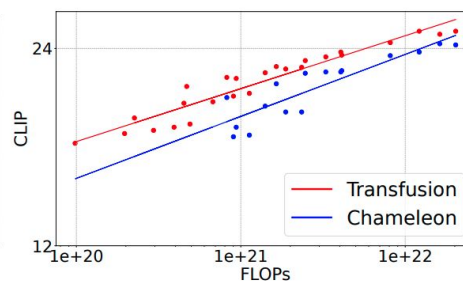
Llama 2 Eval Suite Accuracy



MS-COCO 5k CIDEr



MS-COCO 30k FID



MS-COCO 30k CLIP

Transfusion – Evaluations



Remove the cupcake on the plate.



Write the word "Zebra" in Arial bold.



Change the tomato on the right to a green olive.



Change this to cartoon style.



References

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